

# Lab Exam

## Arpan Adhikari

### Set A

Decoder 3\*8:

Behavioral:

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```
library IEEE;
use IEEE.std_logic_1164.all;

entity arpan_decoder is
port(
    I : in std_logic_vector(2 downto 0);
    Y : out std_logic_vector(7 downto 0)
);
end arpan_decoder;

architecture behav_decoder of arpan_decoder is
begin
    process(I)
    begin
        case I is
            when "000" => Y <= "00000001";
            when "001" => Y <= "00000010";
            when "010" => Y <= "00000100";
            when "011" => Y <= "00001000";
            when "100" => Y <= "00010000";
            when "101" => Y <= "00100000";
            when "110" => Y <= "01000000";
            when "111" => Y <= "10000000";
            when others => Y <= "0";
        end case;
    end process;
end behav_decoder;
```

## Testbench:

```
library IEEE;
use IEEE.std_logic_1164.all;

entity arpan_test_decoder is
end arpan_test_decoder;

architecture test_decoder of arpan_test_decoder is
component arpan_decoder is
port(
    I : in std_logic_vector(2 downto 0);
    Y : out std_logic_vector(7 downto 0)
);
end component;

signal I : std_logic_vector(2 downto 0);
signal Y : std_logic_vector(7 downto 0);

begin
    UUT: arpan_decoder port map
    (
        I => I,
        Y => Y
    );

    stim_proc : process
    begin
        I <= "000"; wait for 100 ps;
        I <= "001"; wait for 100 ps;
        I <= "010"; wait for 100 ps;
        I <= "011"; wait for 100 ps;
        I <= "100"; wait for 100 ps;
        I <= "101"; wait for 100 ps;
        I <= "110"; wait for 100 ps;
        I <= "111"; wait for 100 ps;
    end process;
end test_decoder;
```

## Waveform:



## Mux 4\*1:

### Behavioral:

```
library IEEE;
use IEEE.std_logic_1164.all;

entity arpan_mux is
port(
    I : in std_logic_vector(3 downto 0);
    S : in std_logic_vector(1 downto 0);
    Y : out std_logic
);
end arpan_mux;

architecture behav_mux of arpan_mux is
begin
    process(I)
    begin
        case S is
            when "00" => Y <= I(0);
            when "01" => Y <= I(1);
            when "10" => Y <= I(2);
            when "11" => Y <= I(3);
            when others => Y <= '0';
        end case;
    end process;
end behav_mux;
```

## Testbench:

```
library IEEE;
use IEEE.std_logic_1164.all;

entity arpan_test_mux is
end arpan_test_mux;

architecture test_mux of arpan_test_mux is
  component arpan_mux is
  port(
    I : in std_logic_vector(3 downto 0);
    S : in std_logic_vector(1 downto 0);
    Y : out std_logic
  );
end component;

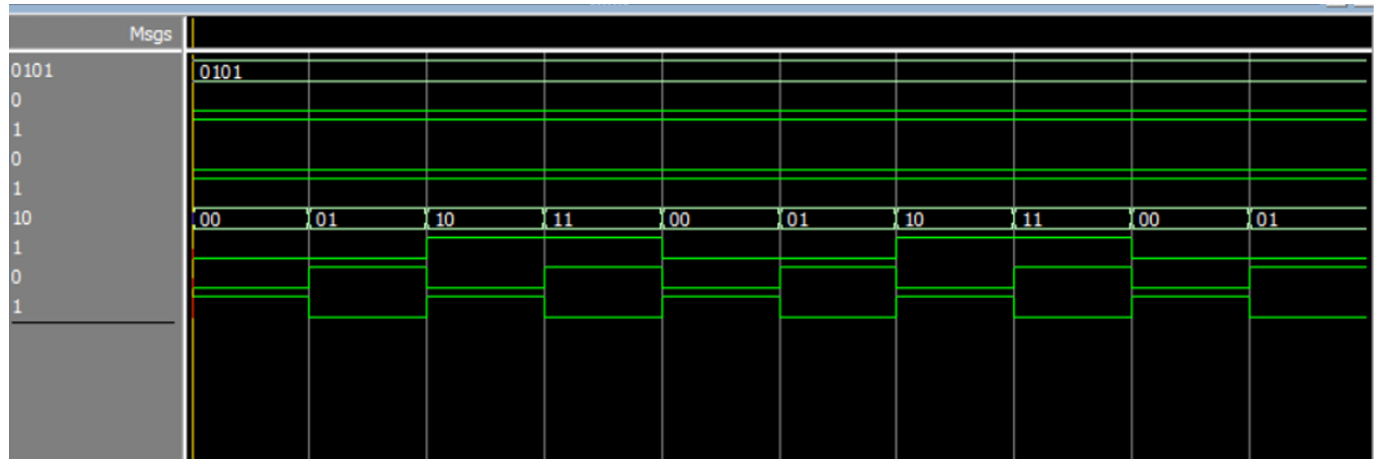
  signal I : std_logic_vector(3 downto 0) := "0101";
  signal S : std_logic_vector(1 downto 0);
  signal Y : std_logic;

begin
  UUT: arpan_mux port map
  (
    I => I,
    S => S,
    Y => Y
  );

  stim_proc : process
  begin

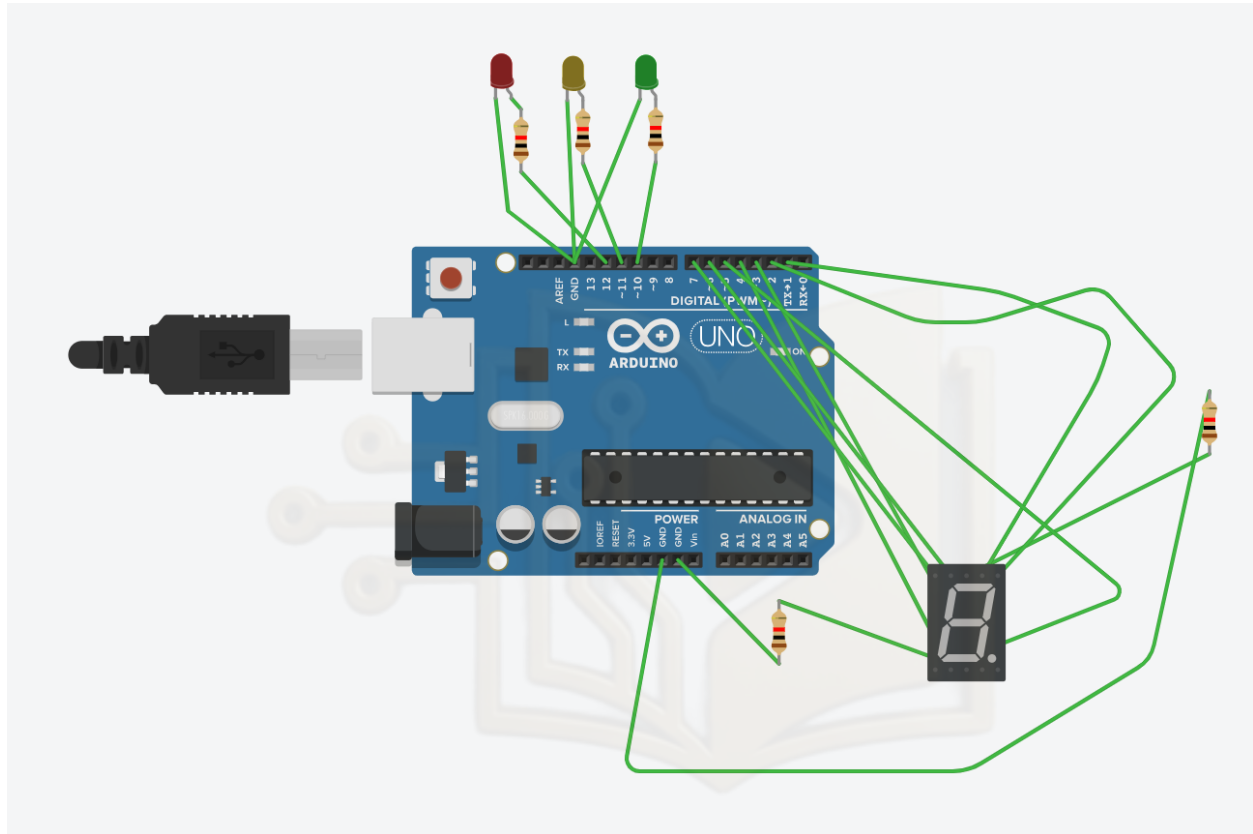
    S <= "00"; wait for 100 ps;
    S <= "01"; wait for 100 ps;
    S <= "10"; wait for 100 ps;
    S <= "11"; wait for 100 ps;
  end process;
end test_mux;
```

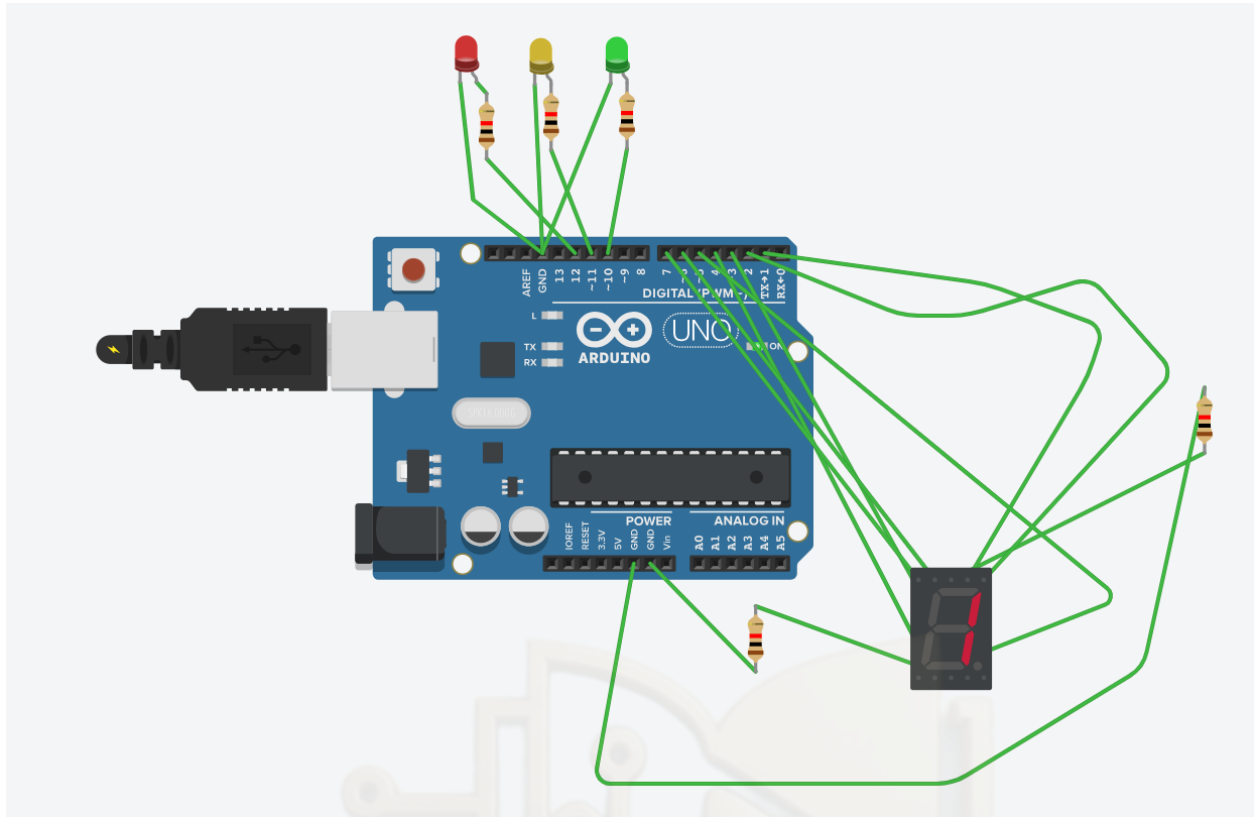
## Waveform:



**Tinkercad:**

**7 segment display with LED traffic light:**





## Ultrasonic Sensor Interfacing to measure distance in meters:

